Chapter 1: Neuroscience: Past, Present, Future

- Prehistoric Fossil: 5000BCE

  "trepanation" 开颅 "milestone for neuroscience" way of cure, not to kill.

- Ancient Egypt: 3000 BCE

  heart is seat of spirit

  mummy, all parts preserved (except the brain)

  record of brain disorder

  heart is vital, heart >> brain.

- Ancient Greece: 400 BCE

  correlation between structure & function.

  nose, ear, eye, tongue all connected to head.

  brain serve the function of sensation

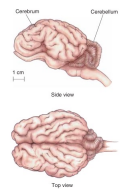
  Hippocrate: "father of western medicine" brain: organ for sensation.

  Aristotle: heart is seat of spirit

  brain serve as heat radiator [cool off blood]

- Ancient Roman Empire. Glad. Gladius. Gladiator 剑 角斗士

  Galen: physician of gladiator



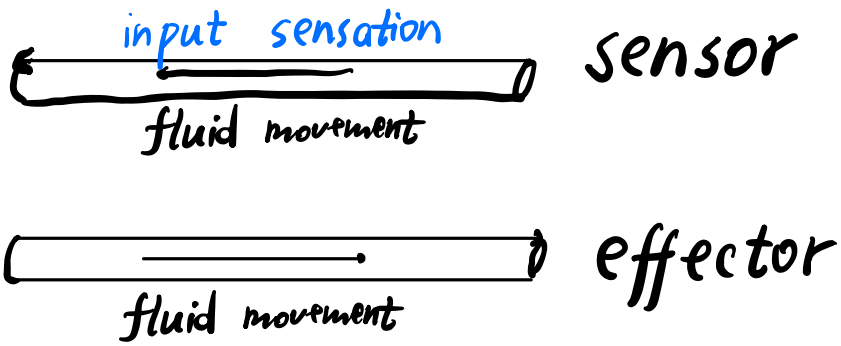
dissection of sheep brain

Parts	Location	Feeling	Function
Cerebrum 大脑	Front	Soft/dough	memory
Cerebellum 小脑	Back	hard	movement
ventricle 脑室	hollow space filled with fluid.		

humorism (体液学说) ⟹ "nerves as tubes"



Brain



Renaissance

Andreas Vesalius father of modern anatomy

reject "humorism", as nerve fiber is solid.

➡ keen for dissection of corpse

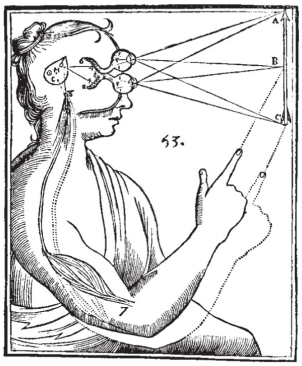
"thief of corpse"

➡ adding details to brain structure.

Descartes: "mind-brain problem"

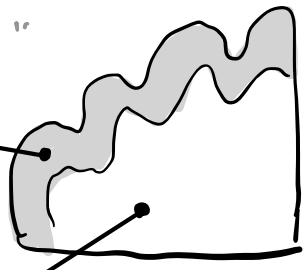
invention: "hydraulic-powered device"

"fluid-mechanism-theory"



1601 ~ 1800.

"Joseph Gall"

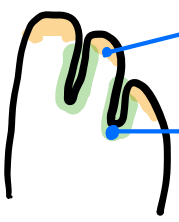


gray matter
site of soma, computing

white matter
continuous → deliver signal [passage]

CNS central nervous system
- brain: cerebrum, cerebellum, brain stem.
- spinal cord:

PNS peripheral nervous system.



bumps → gyri

groove → sulci / fissures

enlarge surface area → intelligence

"lobes" → function localization ?

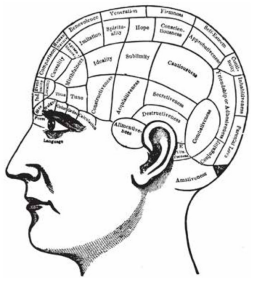
Joseph Gall

shapes of bumps → specific function.

shape of skull ⟷ shape of brain

"Phrenology" 颅相学. falsified

Pseudoscience.



▲ FIGURE 1.10
A phrenological map. According to Gall and his followers, different behavioral traits could be related to the size of different parts of the skull. (Source: Clarke and O'Malley, 1968, Fig. 118.)

Pierre Flouren NYT

experimental ablation method

cerebrum ⟶ sensation & perception.

cerebellum ⟶ movement coordination.

all parts of brain cortex ⟶ same function. "errors"

Benjamin Franklin : observe electrical phenomena

Bois Reymond:

- create device that detect voltage changes
- muscle & brain can generate electricity.

Luigi Galvani: muscle twitch

frog leg twitch when electrical stimulated.

"Volta" ⟶ "voltage"

"Nerves as wires" "electrical signal"

SYW



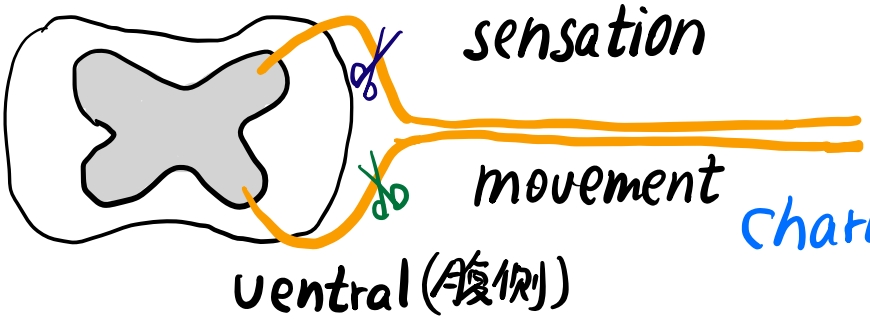
"bidirectional"

sensation

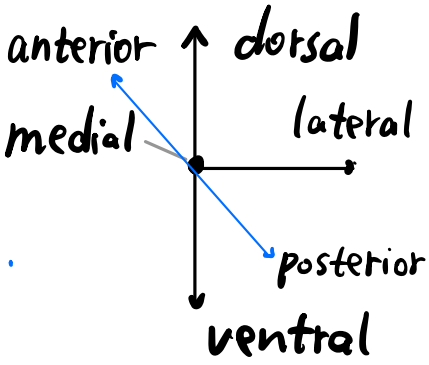
& movement control.

experimental ablation

François Magendie



Charles Bell.



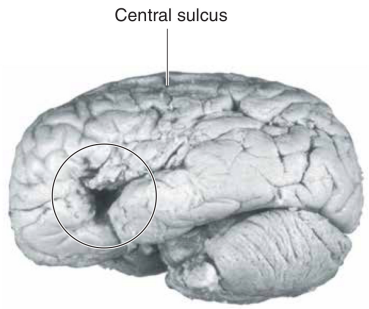
SD MV SMDV

1861.4.17.

"Paul Broca" Patient "TAN". Aphasia 失语症

"cortical function" localization

"frontal lobe" ——→ speech



▲ FIGURE 1.12
The brain that convinced Broca of localization of function in the cerebrum. This is the preserved brain of a patient who had lost the ability to speak before he died in 1861. The lesion that produced this deficit is circled. (Source: Corsi, 1991, Fig. III, 4.)

Gustav Fritsh & Eduard Hitzig.

electrical stimuli on brain cortex on dog

——→ discrete movement

Hermann Munk

experimental ablation occipital lobe ——→ vision

Charles Darwin :

a lot of mammalian share the feature of facing frightening scene.

- Pupil enlarge
- heart beat increase
- hair stand

} fight & flight

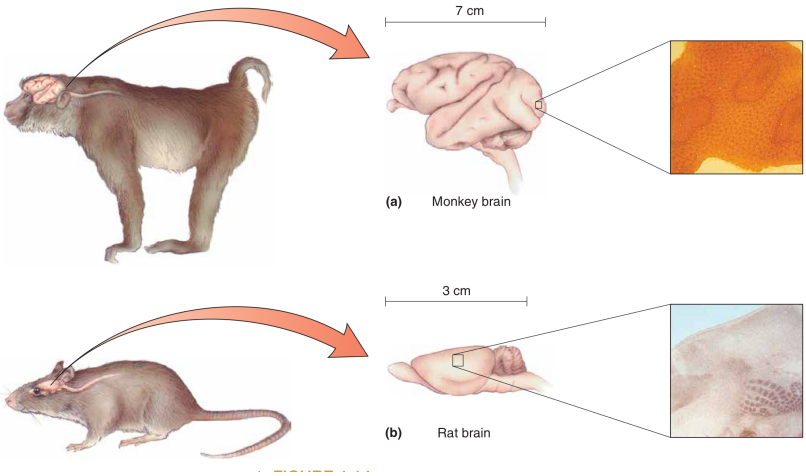
"common ancestry"

adaptation ⟶ environment / niche

relative size of cortex

macaque monkey ⟶ vision

rat / mouse ⟶ somatosensory "whisker"



Major Nervous Disorders

Alzheimer's Disease (AD) +75 years old
progressive degenerative brain disorder.
渐进式 退行性
dementia. fatal

Parkinsons Disease
progressive degenerative brain disorder.
initiating of voluntary movement

Stroke
"failure of **blood supply**", affect specific brain function

autism early on childhood,
psychotic disorder, communication & social,
repetitive behavior

Cerebral Palsy "before, during, soon after birth."
motor cortex disorder,

Schizophrenia, psychotic disorder
delusion, hallucination

Spinal Paralysis traumatic **spinal cord damage**.
feeling & **motor control** disabled

depression: insomnia, loss of appetite. dejection.

Levels of analysis in neuroscience.

- molecular neuroscience.
  e.g. NT: neurotransmitter
  ion channels
- cellular neuroscience.
  types of neuron. "dendritic spin"
- system neuroscience
  perception of signal, e.g. visual, auditory
  motor control. e.g. motor system
- behavioral neuroscience.
  gender differ. addictic drug.
- cognitive neuroscience.
  self-awareness. language.